

From hyperspectral CHRIS to four RGBN bands

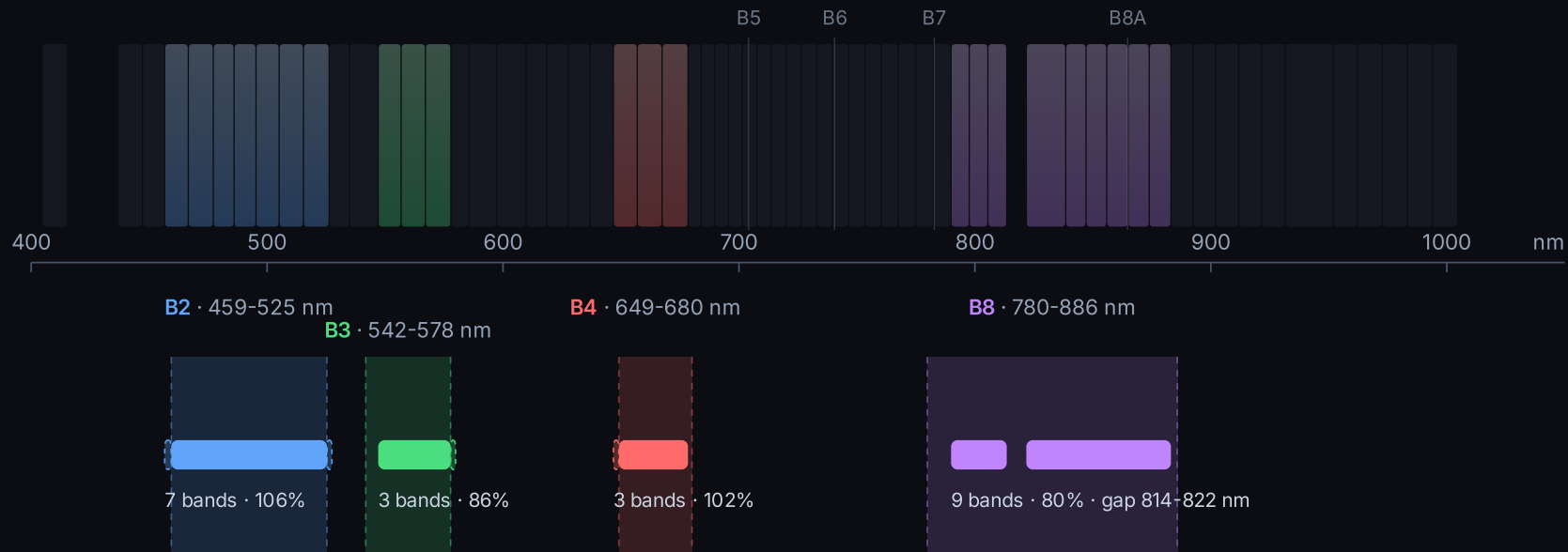
The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

which each CHRIS band goes to the Sentinel-2 band it overlaps the most (max spectral overlap)

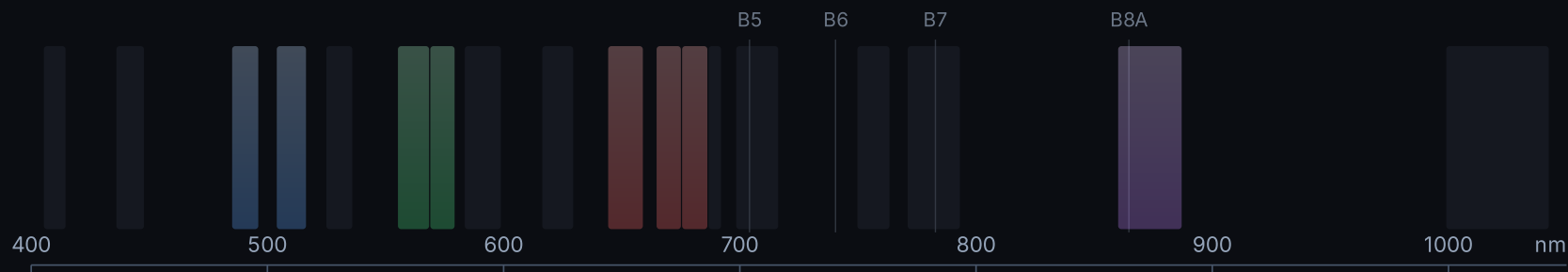
how plain **mean** of the CHRIS bands assigned to each pseudo-band

then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

Mode 1 · 62 bands · ~34 m



Mode 2 · 18 bands · ~17 m

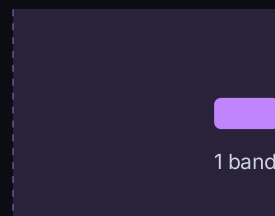
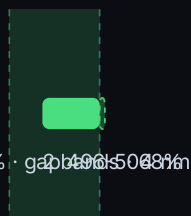
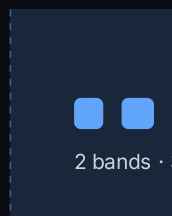


B2 · 459-525 nm

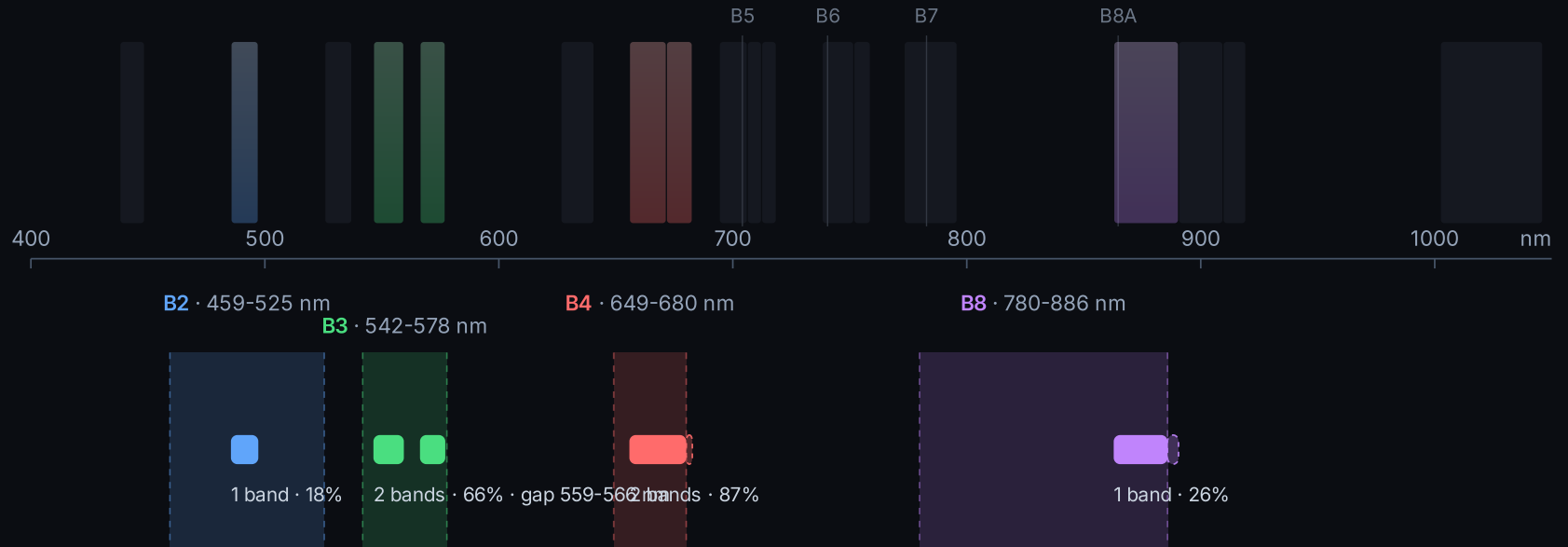
B3 · 542-578 nm

B4 · 649-680 nm

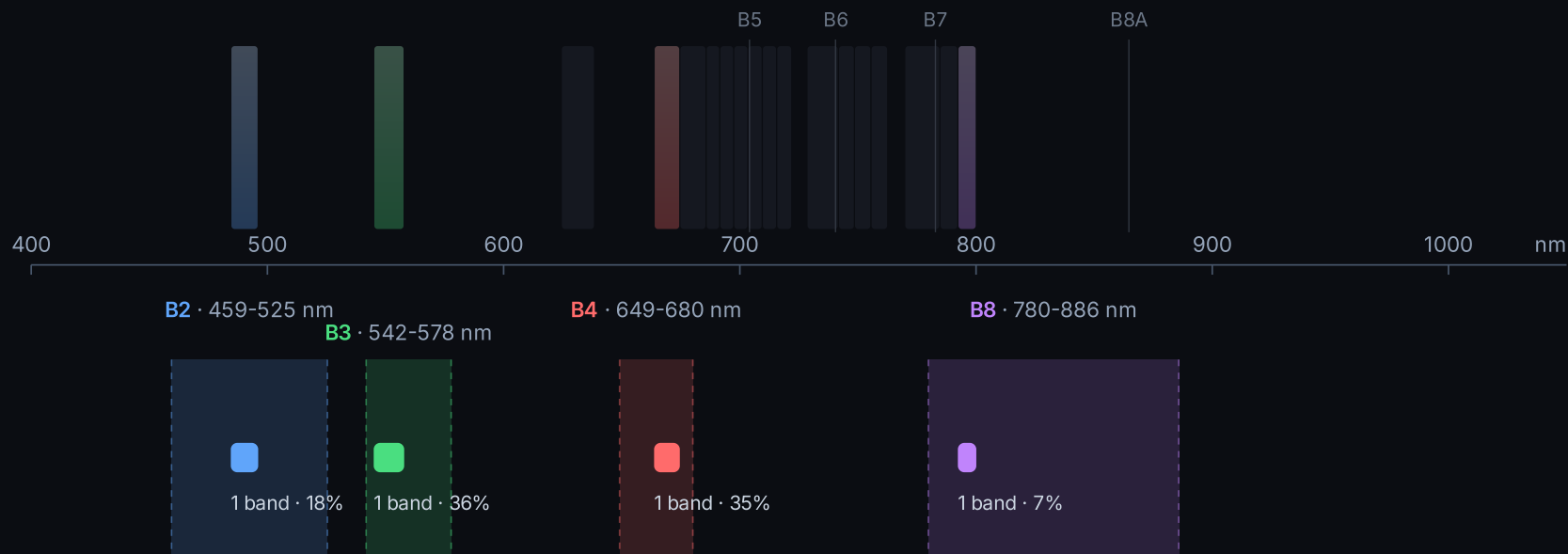
B8 · 780-886 nm



Mode 3 · 18 bands · ~13 m

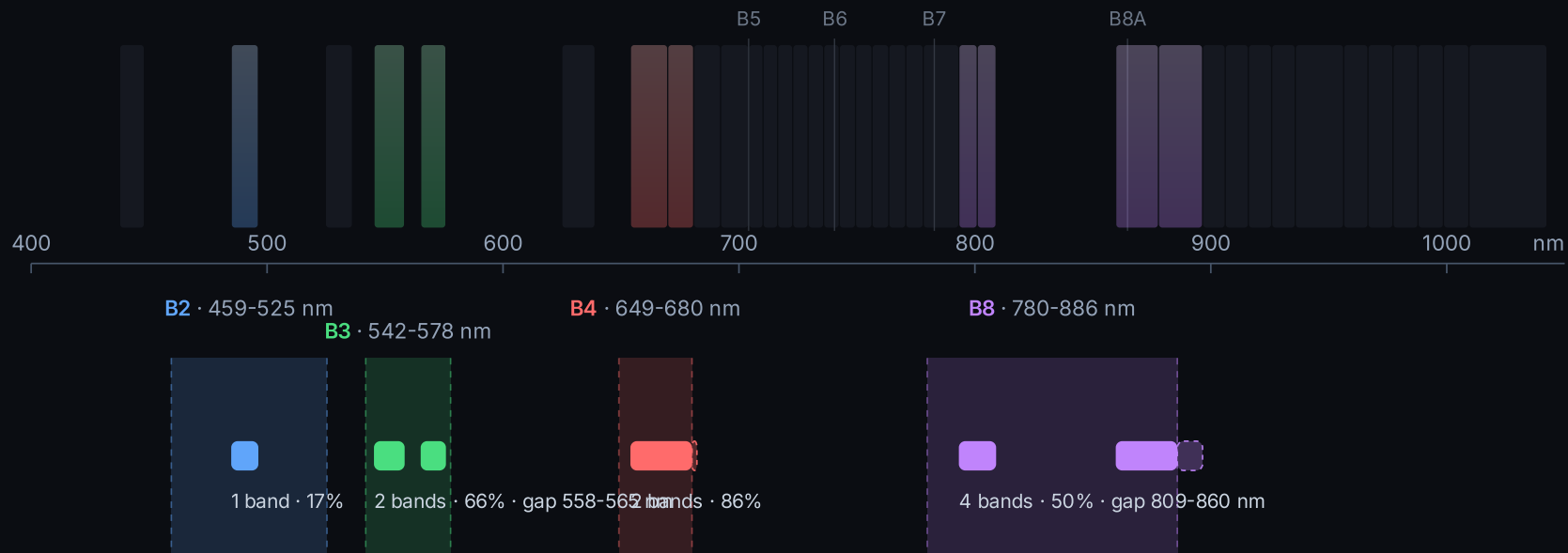


Mode 4 · 18 bands · ~11 m

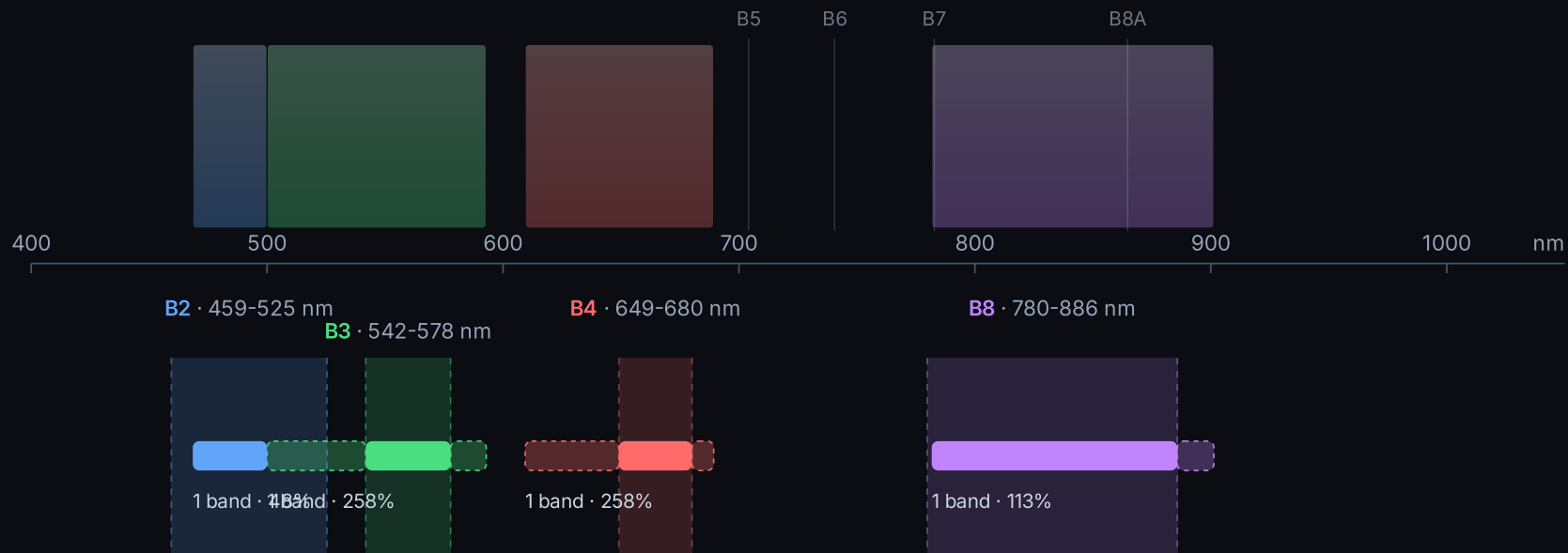


Mode 4 has one clean band per target: 1 → B, 2 → G, 4 → R, 18 → NIR.

Mode 5 · 37 bands · ~20 m



Mode 20 · 4 bands · ~17 m · DN only



Mode 20 has exactly 4 bands: no averaging. Each one is wide and sits shifted from the S2 range it falls into.

Where the bands end up

R · B4

G · B3

B · B2

NIR · B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7) · 457-526 nm	13-15 (3) · 547-578	23-25 (3) · 647-678	43-51 (9) · 790-883
2	3-4 (2) · 485-517	6-7 (2) · 555-580	10-12 (3) · 644-686	17 (1) · 860-887
3	2 (1) · 486-497	4-5 (2) · 546-577	7-8 (2) · 656-683	15 (1) · 863-890
4	1 (1) · 484-496	2 (1) · 545-558	4 (1) · 664-675	18 (1) · 792-800
5	2 (1) · 485-496	4-5 (2) · 545-576	7-8 (2) · 654-681	23-26 (4) · 793-896
20	1	2	3	4

Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order for the model: R, G, B, NIR.